Hesam Nezhadmohammad

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Research Interests

Wireless communications, reconfigurable intelligent surfaces (RIS), beyond-diagonal RIS (BD-RIS), multi-antenna signal processing

Education

Shahid Beheshti University (National University of Iran) — Tehran, Iran
M.Sc. in Electrical Engineering (Communication Systems); GPA: 17.05/20 (3.42/4.0) — *Sep. 2021–Sep. 2024*

- **Thesis:** *Using Reconfigurable Intelligent Surfaces (RISs) for Improvement of Indoor Wireless Communication Environment*
- **Thesis Grade:** *19.5/20 (A+)*
- **Supervisors:** *Dr. Reza Asvadi, Dr. Abbas Pirhadi*

Azad University, Karaj — Karaj, Iran
B.Sc. in Electrical Engineering; GPA: 15.23/20 (3.22/4.0 in the last 60 credits) — *Sep. 2016–Oct. 2020*

- **Thesis:** *Energy Management in Energy-Harvesting Wireless Communication Systems*
- **Thesis Grade:** *17/20 (A)*
- **Supervisor:** *Dr. Farzad Golshan*

Publications

Manuscript in Preparation

1. **H. Nezhadmohammad**, R. Asvadi, and A. Pirhadi, "Outdoor-to-Indoor Spherical-Wave Channel Modeling Using Multi-Panel BD-RIS: Spatial Multiplexing and Inter-Panel Spacing Analysis," *IEEE OJ-COMS*, 2026.

Selected Research Manuscripts

1. **H. Nezhadmohammad**, R. Asvadi, and A. Pirhadi, "Beyond-Diagonal RIS-Aided Outdoor-to-Indoor mmWave Communications," 2025. [PDF] [code]
2. **H. Nezhadmohammad**, R. Asvadi, and A. Pirhadi, "Impact of Beyond-Diagonal RIS Inter-Panel Spacing in mmWave Communications," 2026. [PDF] [code]

Awards & Honors

- **Ranked 2nd** among Master's Graduates in the Department of Electrical Engineering, Shahid Beheshti University
- **Ranked with the Top 0.2%** of Iranian University Entrance Exam for Master's Program in Communication Systems Engineering

Teaching Experience

Shahid Beheshti University (National University of Iran) — Tehran, Iran
Teaching Assistant – Wireless Communications — *Fall 2024*
Instructor: Dr. Reza Asvadi

- Tutored graduate students and provided problem-solving assistance during tutorials.
- Answered students' questions and supported understanding of key wireless communication concepts.
- Graded assignments and exams, and assisted in preparing course projects and exercises.

Academic Projects

- **Graph-Theoretic Visualization of BD-RIS Sparsity Patterns** [code]
  Developed a graph-based framework to construct and visualize the susceptance-matrix sparsity patterns of fully-, group-, single-, tridiagonal-, arrowhead-, band-, and stem-connected BD-RIS architectures, enabling direct comparison of their structural complexity.
- **Two-Stage BD-RIS Beamforming Design Without a Direct BS-User Link** [code] [report]

Extended a two-stage BD-RIS beamforming framework to cascaded BS-RIS-user channels without a direct link, deriving the gradient and descent direction and applying symmetric-unitary projection for fully- and group-connected architectures.

- **Beam-Hopping Spectrum Sharing Between Terrestrial and Non-Terrestrial Networks** [code] [report]
  Explored coordinated spectrum sharing between terrestrial and non-terrestrial networks, evaluating how beam hopping and time synchronization affect terrestrial service capacity under increasing traffic and different timeslot lengths.
- **LEO mmWave and sub-THz Cross-Link Interference** [code] [report]
  Evaluated directional cross-link interference in LEO satellite constellations at mmWave and sub-THz frequencies, examining how constellation density, antenna beamwidth, and orbital geometry affect SINR and achievable capacity.
- **Sudoku Puzzle Solver via Convex Optimization** [code]
  Formulated Sudoku as a binary linear feasibility problem with row, column, box, and clue constraints, and solved it in MATLAB using CVX and MOSEK as part of a graduate convex optimization course.

Selected Courses

Wireless Communications (A+ 4.0/4.0); Channel Coding (A+ 4.0/4.0); Convex Optimization (A 4.0/4.0)

Technical Skills

Programming Languages: MATLAB, Python, C/C++, Lua, Bash
Optimization Methods: Fractional programming, manifold optimization, surrogate optimization, geometric optimization, ADMM, pp-ADMM, SCA, SDR
Frameworks & Libraries: CVX, Manopt, NumPy, SciPy, Pandas, Scikit-learn, PyTorch
Tools & Platforms: Git/GitHub, Linux, Neovim, tmux, Wireless InSite, Inkscape, Blender

Training

Wireless Communications by Prof. Henk Wymeersch; Multiple Antenna Communications by Prof. Emil Björnson; Wireless Communications by Prof. Bruno Clerckx; Convex Optimization for Wireless Communications by Prof. Bruno Clerckx; MATLAB for Scientific Coding by Prof. Mike X Cohen; Python for Scientific Computing by Prof. Mike X Cohen; Statistics and Machine Learning by Prof. Mike X Cohen; A Deep Understanding of Deep Learning by Prof. Mike X Cohen (in progress).

English Language Proficiency

IELTS Academic: Overall 7.0; Speaking 7.5; Writing 7.0
Duolingo English Test: Overall 135; Speaking 145; Writing 140

References

- **Dr. Reza Asvadi**
  Department of Electrical Engineering, Shahid Beheshti University, Tehran, Iran [email]
- **Dr. Abbas Pirhadi**
  Department of Electrical Engineering, Shahid Beheshti University, Tehran, Iran [email]
- **Dr. Esfandiar Mehrshahi**
  Department of Electrical Engineering, Shahid Beheshti University, Tehran, Iran [email]